Correspondence Analysis

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Introduction

Correspondence analysis (CA) is the categorical analogue of principal components analysis. It takes a two-way contingency table and produces a low-dimensional map in which rows and columns become points whose proximity reflects association beyond what would be expected under independence. CA decomposes the chi-squared statistic into row and column contributions across a small number of dimensions, much as PCA decomposes covariance variance, and the resulting biplot is one of the most intuitive ways to visualise the association structure of a contingency table.

Prerequisites

A working understanding of contingency tables, the chi-squared statistic for independence, principal components analysis, and singular value decomposition.

Theory

For an $r \times c$ contingency table with cell counts n_{ij} and margins $n_{i\cdot}$, $n_{\cdot j}$, total n:

1. Compute expected counts under independence $e_{ij} = n_{i\cdot} n_{\cdot j}/n$.
2. Form the matrix of standardised residuals $z_{ij} = (n_{ij} - e_{ij})/\sqrt{e_{ij}}$.
3. Apply singular value decomposition $Z = UDV^\top$.
4. Row coordinates and column coordinates come from U and V scaled by D and the row/column masses; the squared singular values sum to the total inertia χ^2/n.

The biplot places row points close to column points with positive association; points close to the origin are near the average profile and contribute little to the structure. Multiple correspondence analysis (MCA) generalises CA to more than two categorical variables.

Assumptions

Count data in a rectangular contingency table with no zero marginal sums (all rows and columns have at least one observation), and a meaningful chi-squared interpretation of association.

R Implementation

```r
library(FactoMineR); library(factoextra)

data(smoke)
ca_res <- CA(smoke, graph = FALSE)
fviz_ca_biplot(ca_res)
summary(ca_res)
```

Output & Results

`CA()` returns row and column coordinates, contributions to each axis, total inertia, and inertia per axis (the categorical analogue of variance explained). `fviz_ca_biplot()` displays both row and column points on the first two principal axes, with axis labels showing the proportion of inertia captured.

Interpretation

A reporting sentence: "Correspondence analysis of the smoke dataset (5 rank levels × 4 smoking categories) decomposed total inertia of 0.085 into three principal axes; the first axis (88 % of inertia) placed senior employees close to the 'heavy smoker' category and junior managers near 'none', suggesting an age-rank-smoking association beyond chance ($\chi^2 = 16.4$, $p = 0.17$ — the visual association is strong but the statistical test is underpowered for this small table)." Always quote the proportion of inertia displayed and complement with the chi-squared test.

Practical Tips

- Report the proportion of inertia retained in the displayed dimensions; if the first two axes capture less than 70 %, much of the structure is not visible in the biplot.
- For more than two categorical variables, switch to multiple correspondence analysis (MCA, `FactoMineR::MCA`); supplementary continuous covariates can be projected on top.
- Remove or merge zero-count rows or columns before fitting; they break the chi-squared transformation and produce undefined coordinates.
- The symmetric biplot scales rows and columns comparably; the row-principal or column-principal alternatives are useful when one dimension is the focus but distort the joint geometry.
- Pair CA with a chi-squared test on the original table; the inertia decomposition shows where the association concentrates, while the test quantifies whether it exceeds chance.
- For supplementary categories that should not influence the axes (e.g. demographic markers on the original survey items), pass them through the `quali.sup` argument; they are projected after fitting but do not affect axis estimation.

R Packages Used

`FactoMineR` for `CA()`, `MCA()`, and the broader factor-analysis family; `factoextra` for `fviz_ca_biplot()`, `fviz_ca_row()`, `fviz_ca_col()` ggplot-style visualisation; `ca` for an alternative implementation with classic biplot styling; `ade4::dudi.coa()` for ecology-flavoured CA.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- PCA, factor analysis, CCA, LDA
- Clustering: k-means, hierarchical, model-based
- UMAP and t-SNE